FERDOUS WAHID ANIK

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RESEARCH INTERESTS

Artificial Intelligence, Clinical NLP, Large Language Model, Biomedical informatics, Biomedical Signal Processing.

EDUCATION

Bachelor of Science in Biomedical Engineering **February 2019 - February 2023**

Military Institute of Science and Technology (MIST), Dhaka. Degree awarded by Bangladesh University of Professionals (BUP).

- CGPA 3.30 / 4.00. Degree conferred April 2023.
- Thesis: "Design and Development of a Portable SCG-ECG Dual Signal Acquisition System."
  Supervisor: Dr. Md. Asadur Rahman. Grade: A+ (4.00 / 4.00).
- Coursework: Digital Signal Processing, Biomedical Instrumentation, Bioinformatics, Medical Imaging, Probability and Statistics, Computer Programming.

RESEARCH EXPERIENCE

Research Associate and Technical Lead **September 2025 -Present**

World Bank-funded Higher Education Acceleration & Transformation (HEAT) Project, Department of Pharmacy & Institute of Research, Innovation, Incubation and Commercialization (IRIIC), United International University, Dhaka.

- Prepared the technical proposal, system architecture, methodology and budget for the HEAT-funded project.
- Organized consultative workshop with stakeholders to refine the protocol and requirements for developing healthcare solutions.
- Develop evidence-retrieval, clinical-knowledge and language-model components for prescription review and decision support.
- Support project administration through procurement, budget follow-up, reporting and interdisciplinary communication.

Research Engineer **July 2023 -August 2025**

Advanced Intelligent Multidisciplinary Systems (AIMS) Lab, IRIIC, United International University, Dhaka.

- Conduct clinical-AI and biomedical-informatics research on medication safety and prescription auditing.
- Design and curate structured clinical datasets from medical guidelines, literature and expert consultation.
- Presented the clinical use and implementation plan for our prescription-support work in the Innovation Cohort Program.
- Develop data-processing workflows and knowledge-based methods for medication-error assessment.
- Support research design, literature review, technical documentation, manuscript preparation and system evaluation.
- Supervise undergraduate research activities and mentor interns in data engineering and research practice.

Undergraduate Thesis Researcher **March 2022 - February 2023**

Department of Biomedical Engineering, MIST, Dhaka.

- Developed a portable ECG–SCG acquisition system for synchronized monitoring of cardiac electrical and mechanical activity.
- Built MATLAB-based signal-processing workflows by denoising, fiducial detection and filter designing .
- Collected and analysed paired ECG–SCG recordings from ten volunteers, evaluated heart-rate estimates against ECG and led the drafting and revision of the first-author manuscript now under review at Journal of Sensors.

Undergraduate Researcher, Final Year Design Project **March 2022 - February 2023**

Department of Biomedical Engineering, MIST, Dhaka.

- Compared alternative imputation, normalization, sampling and feature-engineering methods for gestational diabetes prediction, evaluating how preprocessing choices affected intermediate results and downstream model performance. Coauthored the resulting ICCIT 2023 paper.

PUBLICATIONS

Peer-Reviewed Journal Article

[J1] **Anik, F. W.**, Zaman, M., Foyez, T. and Mamun, K. A. "DOSAGE: Dataset for Optimized Safe Antibiotic Guidelines and Estimates." *Scientific Data, Nature* 12, 1662 (2025). doi.org/10.1038/s41597-025-05761-8

Peer-Reviewed Conference Proceedings

[C1] **Anik, F. W.**, Al Rifat, F., Zaman, M., Sarker, F. and Mamun, K. A. "The Importance of Prescription Audits: Navigating Complexities and AI-Driven Advancement." *2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, 2024. **doi.org/10.1109/BECITHCON64160.2024.10962770**

[C2] Riyed, T. H., Nabi, T., Dutta, A., Hasan, M. K., **Anik, F. W.** and Ornob, A. "Gestational Diabetes Prediction in Pregnancy: A Machine Learning and Data Preprocessing Approach." *26th International Conference on Computer and Information Technology (ICCIT)*, 2023. **doi.org/10.1109/ICCIT60459.2023.10441112**

Published Dataset

[D1] **Anik, F. W.**, Zaman, M., Foyez, T. and Mamun, K. A. "DOSAGE: Personalized Antibiotic Medication Dataset." Figshare, 2025. **doi.org/10.6084/m9.figshare.28692437.v1**

Manuscripts Under Review

[U1] **Anik, F. W.**, Ahmed, F., Al Rifat, F., Zaman, M., Sarker, F. and Mamun, K. A. "From Error to Accuracy: A Systematic Review of Automated Approaches to Detect and Correct Prescription Errors in Health Services." *BMC Health Services Research*, submitted 2025.

[U2] Shahid, R., **Anik, F. W.**, Das, R., Foyez, T., Ohidujjaman and Mamun, K. A. "STABLE: A Structured Dataset for Cardiovascular Drug Prescription Auditing." *Scientific Data*, *Nature* 2026.

[U3] **Anik, F. W.**, Al Rifat, F., Zaman, M., Sarker, F. and Mamun, K. A. "Evaluation of a Knowledge-Driven System for Detecting Antibiotic Dosage Errors in Prescriptions." Array, Elsevier, submitted 2025.

[U4] **Anik, F. W.**, Chowdhury, M. S. and Rahman, M. A. "Design and Development of a Portable SCG-ECG Dual Signal Acquisition System." *Journal of Sensors, Wiley* , submitted 2026.

[U5] Sarker, F., **Anik, F. W.**, Ahmed, F., Hossen, M. R., Zaman, M. and Mamun, K. A. "Feasibility and User Acceptance of a Digital Workplace Wellness Program for NCD Risk Management in a LMIC: A RE-AIM Framework Analysis." BMC Public Health, submitted 2025.

[U6] **Anik, F. W.**, Salim, M. B., Shahid, R., Foyez, T., Ohidujjaman and Mamun, K. A. "A Knowledge-Constrained Architecture for Automated Prescription Auditing." Submitted in 29th International Conference on Computer and Information Technology, 2026.

[U7] Salim, M. B., **Anik, F. W.**, Shahid, R., Foyez, T., Ohidujjaman and Mamun, K. A. "Multi-LLM Benchmarking for Prescription Error Detection: A Clinical Evaluation Framework on Real-World Outpatient Data." Submitted to the *7th International Conference on Electrical Engineering and Information & Communication Technology*, 2026.

RESEARCH GRANTS & PROPOSALS

World Bank Academic Transformation Fund, Higher Education Acceleration & Transformation (HEAT) Project, University Grants Commission of Bangladesh **2025**

Awarded BDT 45.3 million (about USD 377,000). Principal Investigator: Dr. Tahmina Foyez.

- Full title: "SmartAMR: Advancing Digital Healthcare with AI-driven Intelligence to Strengthen Medical Adherence, Minimize Prescription Errors, and Reduce Antimicrobial Resistance in Bangladesh."

Institute for Advanced Research, United International University **2023**

Awarded BDT 0.5 million (about USD 4,100). Principal Investigator: Prof. Khondaker A. Mamun, PhD.

- Full title: "Feasibility of Prescription Audit and Methodology Development"

Additional proposals prepared and submitted*: MIT Solve x The Trinity Challenge (2025); Research and Innovation Center, Bangladesh (2024).*

CONFERENCE PRESENTATIONS

Accepted oral abstract: "A Knowledge-Driven Study for Identifying Anomalous Antibiotic Prescriptions."

Global Public Health Symposium 2025, Colombo, Sri Lanka, July 2025. Presentation delivered by my supervising professor Dr. (Khondaker A. Mamun).

Scientific Poster: "Feasibility of Automated Prescription Audit Systems for Better Patient Care"

International Health Economics Conference 2023: Progress Towards Universal Health Coverage.

TEACHING AND RESEARCH SUPERVISION

Research Supervisor, Final Year Design Project **December 2023 - June 2024**

Department of Computer Science and Engineering, United International University, Dhaka.

- Supervised an undergraduate team building a data-driven diabetes medication optimization system. Set the architecture and the clinical decision logic the model had to satisfy, then reviewed the finished build against the governing guidelines.

TECHNICAL AND RESEARCH SKILLS

Programming and Data: Python (NumPy, Pandas, scikit-learn, NLTK, OpenCV), MATLAB, R, C.

Machine Learning and AI: PyTorch, TensorFlow, RAG pipeline development, prompt engineering, LLM fine-tuning (LoRA).

Research Methods: Systematic review, dataset curation and validation, clinical knowledge representation, expert-panel protocol design, benchmark evaluation.

Signal Processing and Instrumentation: Digital filtering and denoising, biomedical sensor integration, ECG and seismocardiography acquisition, embedded prototyping.

Tools: LaTeX, Git, OriginPro.

Languages: English (B2) IELTS Academic overall 6.5, with no component below 6.0.

LEADERSHIP AND SERVICE

President, MIST Einthoven Club **2021 - 2022**

Department of Biomedical Engineering, MIST, Dhaka.

- Directed ANURONON'22, the first national biomedical engineering event hosted by MIST, which drew teams from universities and colleges across Bangladesh.

General Secretary, IEEE MIST Student Branch **2021 - 2022**

Institute of Electrical and Electronics Engineers, MIST Student Branch, Dhaka.

- Programmed the branch calendar of guest lectures, technical competitions and workshops, coordinating speakers and teams.

Administrative Lead, Team SPARK **2020 - 2021**

IEEE Computer Society Bangladesh Chapter.

Additional roles: General Secretary, MIST Literature and Culture Club (2021 - 2022); Class Representative, Department of Biomedical Engineering.

REFERENCES

Dr. Md. Asadur Rahman

Associate Professor & Chairperson of the Department of Artificial Intelligence and Data Science (ADS),
Green University of Bangladesh (GUB), Dhaka, Bangladesh
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Dr. Khondaker A. Mamun

Director, AIMS Lab & IRIIC; Professor, Department of Computer Science and Engineering
United International University, Dhaka, Bangladesh
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Dr. Tahmina Foyez

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